

MODULE 06

Evaluation of GenAI Systems: Accuracy, Faithfulness, and Trust

This module focuses on building robust evaluation frameworks for generative AI systems. We'll explore how to measure accuracy, ensure faithfulness to source material, and establish trust through systematic testing methodologies for chatbots and RAG (Retrieval-Augmented Generation) systems.

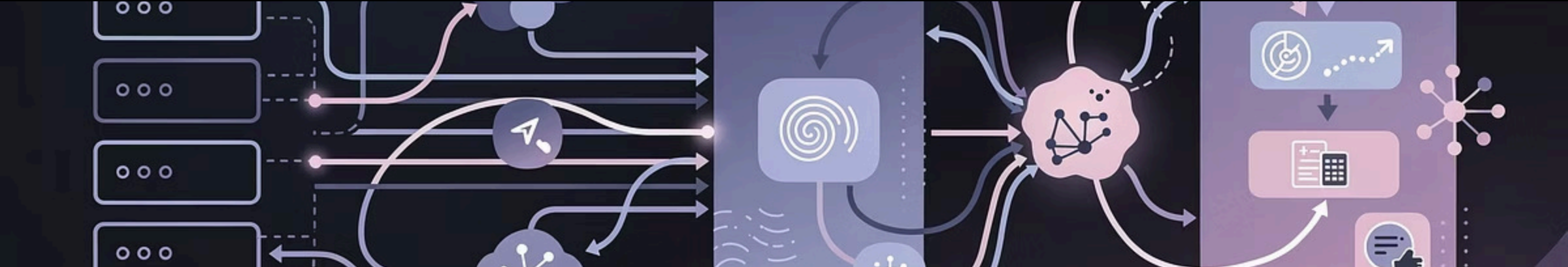
GENAI & NLP

PROMPT ENGINEERING

RAG SYSTEMS

GENAI, NLP AND PROMPT ENGINEERING





Recap: What We've Built So Far

01

Retrieval Mechanism

We implemented a retrieval system that finds relevant chunks from our knowledge base, forming the foundation of our RAG pipeline.

02

Prompt Grounding

We added strict instructions to force the model to use only provided sources, establishing boundaries for response generation.

03

RAG with Citations

We built a complete RAG system that answers with proper citations and includes no-answer fallback when information isn't available.

04

Quality Verification

Now we must verify that our system produces correct, faithful, safe, and consistent outputs across diverse queries.

Module Overview and Learning Outcomes



This module equips you with practical skills to evaluate GenAI and RAG systems systematically. You'll learn to create benchmark datasets, measure retrieval quality, and assess answer faithfulness using both automated metrics and human evaluation frameworks.

Key Skills You'll Master

- Understanding evaluation principles for GenAI and RAG systems
- Building gold standard test datasets with ground truth answers
- Measuring retrieval accuracy using Precision@K metrics
- Grading answers with faithfulness and helpfulness rubrics
- Maintaining comprehensive evaluation logs in spreadsheets

What is Evaluation and Why It Matters

Evaluation transforms subjective quality assessment into measurable, repeatable processes. For GenAI systems, evaluation is the critical bridge between experimental prototypes and production-ready applications that users can trust.

Accuracy & Grounding

Evaluation verifies that answers are factually correct and properly grounded in source material, ensuring reliability.

Hallucination Detection

Systematic testing reveals when models generate plausible but false information, a critical failure mode in RAG systems.

Retrieval Quality

Evaluation exposes weak retrieval that provides irrelevant or incomplete context, directly impacting answer quality.

Measurable Progress

Quantitative metrics enable systematic improvement cycles rather than relying on intuition or guesswork.

Real-World Example: Return Policy Query

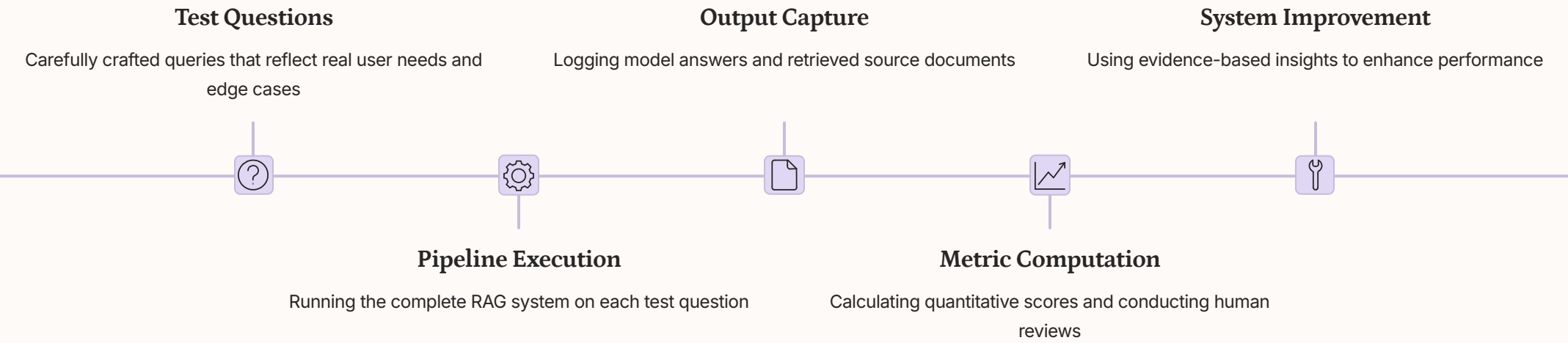
Question: "What's your return policy for electronics?"

Without Evaluation: Model might confidently state "30-day returns" when policy is actually 14 days.

With Evaluation: Testing catches the error before deployment, preventing customer service issues.

Evaluation Components: What We Measure

A comprehensive evaluation framework consists of interconnected components that work together to assess system performance. Each component serves a specific purpose in the evaluation pipeline, from input design to actionable insights.



Component	Purpose	Output Type
Test Set	Provides consistent benchmark for measuring system behavior	Questions + Gold Answers
Retrieval	Assesses quality of context provided to the model	Top-K chunks + scores
Generation	Evaluates final answer quality and citation accuracy	Text + source references
Scoring	Quantifies performance across multiple dimensions	Metrics + rubric scores
Analysis	Identifies patterns, failure modes, and improvement areas	Error categories + fixes

Evaluation Workflow: End-to-End Process

Systematic evaluation follows a structured workflow that transforms subjective quality questions into actionable improvement strategies. This iterative process ensures continuous enhancement of your GenAI system.

1

Create Test Set

Develop questions with verified gold answers that represent real usage patterns

2

Execute System

Run the complete pipeline and capture all outputs for analysis

3

Score Quality

Measure retrieval relevance and answer quality using defined metrics

4

Analyze Errors

Categorize failure modes and identify root causes

5

Implement Fixes

Apply targeted improvements based on evidence

6

Re-Test

Validate improvements with the same test set

"Evaluation isn't a one-time activity—it's a continuous feedback loop that drives systematic improvement. Each iteration teaches you something new about your system's behavior and capabilities."

Creating Gold Test Datasets: Ground Truth

A gold test dataset serves as your evaluation benchmark—a carefully curated set of questions paired with verified, expert-approved answers. These ground truth examples provide the standard against which all system outputs are measured.

Why Gold Datasets Matter

- **Consistency:** Everyone evaluates against the same standard, eliminating subjectivity
- **Hallucination Detection:** Comparing outputs to verified answers reveals fabricated information
- **Coverage Analysis:** Identifies knowledge gaps where the system lacks adequate information
- **Benchmark Stability:** Enables meaningful before/after comparisons across system versions

Recommended Dataset Sizes

For laboratory exercises and initial testing, 8–10 high-quality questions provide meaningful insights. For robust production evaluation, aim for 25–30 questions covering diverse scenarios, edge cases, and common failure modes.

Question	Gold Answer
What is the return policy for electronics?	Electronics can be returned within 14 days of purchase with original packaging and receipt. Opened software cannot be returned.
Do you offer international shipping?	Yes, we ship to over 50 countries. Shipping costs and delivery times vary by destination. See our shipping page for details.

Precision@K: Retrieval Accuracy Metric

Precision@K measures how many of the top-K retrieved chunks are actually relevant to the user's question. This metric directly indicates whether your retrieval system is providing useful context to the language model.

Understanding the Metric

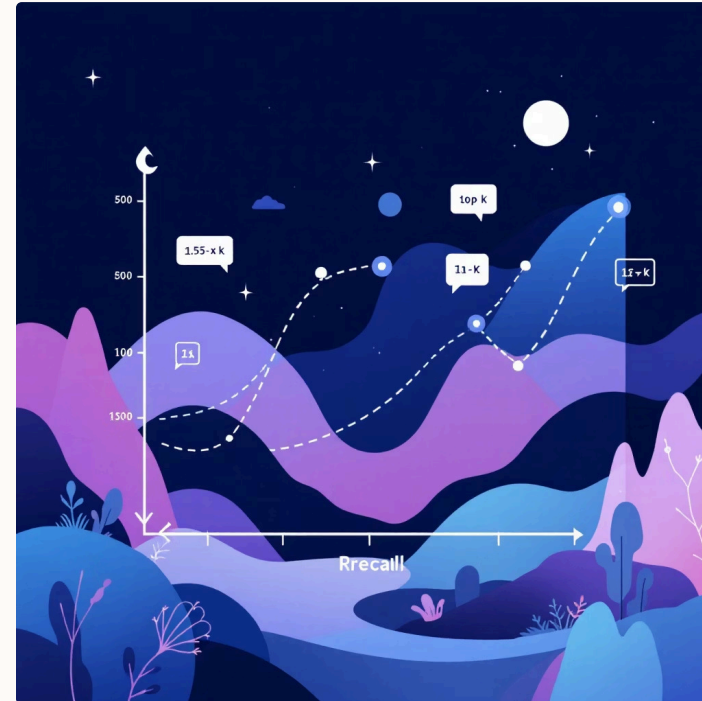
The formula is elegantly simple:

$$\text{Precision@K} = \frac{\text{Number of Relevant Chunks in Top-K}}{K}$$

For example, if you retrieve the top 3 chunks and 2 of them are relevant to answering the question, then Precision@3 = 2/3 = 0.67 or 67%.

Impact on RAG Performance

- **High Precision@K (≥ 0.80):** Strong grounding, relevant context, accurate answers
- **Medium Precision@K (0.50–0.79):** Mixed results, some noise, inconsistent quality
- **Low Precision@K (≤ 0.49):** Weak retrieval leading to poor or hallucinated answers



Real Example

Query: "What are your business hours?"

Top 3 Retrieved Chunks:

- ✓ Store hours: Mon-Fri 9am-7pm
- ✓ Weekend hours: Sat-Sun 10am-5pm
- ✗ Employee parking guidelines

Precision@3 = 2/3 = 0.67

The irrelevant chunk about parking dilutes context quality and may confuse the model.

Faithfulness vs Helpfulness: Two Key Quality Axes

Answer quality exists on two critical dimensions that must both be satisfied. A response that excels on one axis but fails on the other cannot be considered high-quality output from a RAG system.

Faithfulness

Faithfulness measures whether the answer is fully supported by and grounded in the retrieved sources. Every claim, fact, or statement must trace back to the provided context—nothing should be added from the model's training data or imagination.

High Faithfulness Example: "According to our return policy document, electronics can be returned within 14 days with receipt."

Low Faithfulness Example: "Electronics can be returned within 30 days" (when sources say 14 days)

Helpfulness

Helpfulness evaluates whether the answer is clear, complete, and useful to the user. Even if technically accurate, an answer that's confusing, incomplete, or poorly structured fails the helpfulness criterion.

High Helpfulness Example: Clear structure, addresses the question directly, provides relevant details

Low Helpfulness Example: Vague, incomplete, uses jargon without explanation, or misses the question's intent

	Faithful	Unfaithful
Helpful	✓ Ideal Answer — Accurate, clear, and complete	× Dangerous — Clear but contains hallucinations
Unhelpful	△ Needs Work — Accurate but unclear or incomplete	× × Critical Failure — Wrong and confusing

Critical Insight: A helpful but unfaithful answer is particularly risky because it sounds authoritative and useful while containing fabricated information. This is the classic hallucination problem in GenAI systems.

Human Evaluation Rubric: 0–10 Scorecard

Human evaluation provides nuanced assessment that automated metrics cannot capture. This structured rubric breaks down answer quality into five measurable dimensions, each contributing equally to a total score out of 10.

	Faithfulness (0–2 points) All claims supported by sources, no hallucinations or unsupported additions
	Relevance (0–2 points) Directly addresses the user’s question without tangential information
	Clarity (0–2 points) Well-structured, easy to understand, appropriate detail level
	Safety Compliance (0–2 points) Respects privacy, doesn't share restricted data, follows ethical guidelines
	Citation Correctness (0–2 points) Accurate source references, proper attribution when required

Dimension	2 Points
Faithfulness	Fully grounded, no invented facts
Relevance	Directly answers question
Clarity	Clear structure and language
Safety	No policy violations
Citations	Accurate source references

Total Score	Interpretation
9–10	Excellent, production-ready
7–8	Good, minor improvements needed
5–6	Acceptable, significant gaps
3–4	Poor, major issues present
0–2	Unacceptable, critical failures

Spreadsheet-Based Evaluation Logs: Simple and Powerful

Spreadsheets provide an accessible yet powerful tool for tracking evaluation results. They enable transparency, facilitate debugging, and create a historical record of system improvements across versions.

Essential Fields for Your Evaluation Log

1

Query

The exact test question

2

Gold Answer

Expert-verified correct response

3

Model Answer

System-generated response

4

Faithful

Score 0–2 for source grounding

5

Helpful

Score 0–2 for clarity/completeness

6

Precision@3

Retrieval accuracy metric

7

Notes

Observations, error types, fix suggestions

Query	Gold Answer	Model Answer	Faithful	P@3	Total	Notes
Return policy?	14 days with receipt	You can return items within 14 days...	2	1.0	9	Perfect retrieval
Ship internationally?	Yes, 50 countries	We ship worldwide to most countries...	1	0.67	7	Added "worldwide"—not in sources
Business hours?	M-F 9-7, Sat-Sun 10-5	We're open during standard business hours	0	0.33	4	Vague, poor retrieval

Error Analysis: Finding the Root Cause

Systematic error categorization transforms individual failures into actionable insights. By identifying patterns in how and why your system fails, you can implement targeted fixes rather than making random adjustments.

Common Error Types



Hallucinated Facts

Model adds plausible but unsupported information not present in sources



Partial Retrieval

Missing key context chunks that contain the complete answer



Poor Explanation

Factually correct but confusing, overly technical, or incomplete response



Safety Failures

Reveals restricted data or violates ethical guidelines

Error Type	Root Cause
Hallucination	Weak grounding instructions, model over-reliance on training data
Partial Retrieval	Poor chunking, missing content, inadequate keyword coverage
Poor Clarity	Prompt doesn't emphasize readability, lacks formatting instructions
Safety Violation	Missing safety filters, unclear content boundaries
Citation Errors	Prompt doesn't enforce source tracking, weak citation format

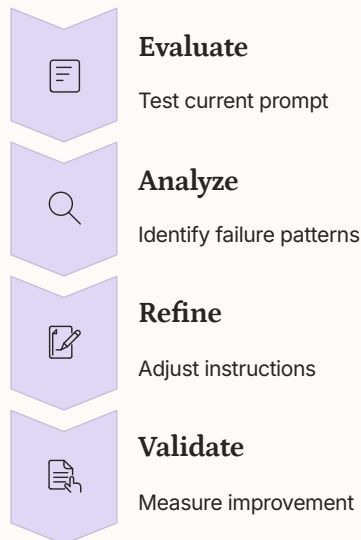
Quick Fix Strategies

- **For Hallucinations:** Strengthen prompt grounding, add "use only sources" emphasis, implement explicit refusal patterns
- **For Retrieval Issues:** Improve chunking strategy, add missing content, enhance metadata
- **For Clarity Problems:** Add structure requirements, provide response templates, emphasize audience needs

Improving RAG via Feedback Loops

Evaluation insights drive two parallel optimization loops that work together to enhance system performance. Each loop targets different aspects of the RAG pipeline, creating a comprehensive improvement framework.

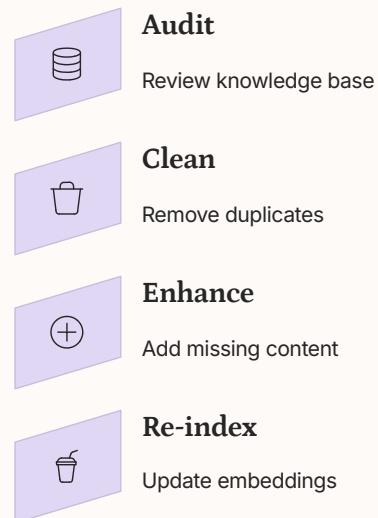
Prompt Optimization Loop



Prompt Optimizations Include:

- Tightening grounding instructions with explicit examples
- Enhancing citation requirements and format
- Adding refusal patterns for out-of-scope queries
- Improving clarity and structure directives

Data Optimization Loop



Data Optimizations Include:

- Removing outdated or duplicate chunks
- Improving content coverage for common queries
- Enhancing chunking strategy for better retrieval
- Adding metadata for improved filtering

Best Practice: Re-evaluate with the same test set after each optimization iteration. This enables you to measure the actual impact of changes and raises both Precision@K and faithfulness scores systematically.

Case Study: E-commerce Support Bot Evaluation

This real-world example demonstrates how systematic evaluation and targeted improvements transform a mediocre RAG system into a reliable production tool.

Initial Evaluation Results

0.68

Precision@3

Inconsistent retrieval quality

7/10

Faithfulness Score

Occasional unsupported details

42%

Perfect Answers

Only 5 of 12 test cases scored 9+

Problems Identified

- **Irrelevant Retrieval:** Old policy versions and unrelated documents polluted top results
- **Extra Details:** Model occasionally added helpful-sounding but unsupported information
- **Inconsistent Citations:** Some answers lacked proper source references
- **Duplicate Content:** Same information appeared in multiple chunks, wasting retrieval slots

Targeted Fixes Implemented

- **Data Cleanup:** Removed duplicates and outdated policy documents
- **Prompt Strengthening:** Added explicit refusal rules and citation requirements
- **Content Updates:** Refreshed policy information and improved chunking
- **Better Metadata:** Enhanced document tagging for improved filtering

Metric	Before Optimization	After Optimization
Precision@3	0.68 (Inconsistent)	0.87 (Reliable)
Faithfulness Score	7/10 (Good)	9/10 (Excellent)
Perfect Answers (9–10)	42% (5 of 12)	83% (10 of 12)
Hallucination Rate	25% (3 cases)	0% (Zero cases)

The improvement cycle took three iterations over two weeks, demonstrating that systematic evaluation drives measurable progress.

Debrief + Homework

Debrief Discussion

→ Trust Through Evaluation

Why systematic evaluation is essential for GenAI system reliability and user trust

→ Precision@3 Insights

What retrieval metrics reveal about context quality and answer grounding

→ Quality Dimensions

Understanding the critical difference between faithful and helpful answers

→ Real Error Example

Discussing one concrete error from testing and the fix that resolved it

Homework Assignment

Complete a hands-on evaluation exercise that applies all concepts from this module:

01

Create Test Set

Prepare **8 test questions** with verified gold answers relevant to your domain

02

Run & Record Retrieval

Execute retrieval for each query and document **Top-3 chunks + Precision@3** score

03

Grade Each Answer

Apply the human evaluation rubric to score each response (**0–10 scale**)

04

Categorize Errors

Tag one error type per question: hallucination / partial retrieval / clarity / safety

Thank You

You've mastered GenAI evaluation fundamentals

You now possess the skills to systematically evaluate RAG systems, measure retrieval quality, assess answer faithfulness, and drive continuous improvement through evidence-based optimization loops.

» COMING NEXT

MODULE 07 – Chatbots: State, Memory & Tools

Building on evaluation foundations, we'll explore how to create stateful conversational agents that maintain context across turns, remember user preferences, and integrate external tools for enhanced capabilities.

Conversation State

Managing multi-turn dialogue context

Memory Systems

Long-term user preference storage

Tool Integration

Connecting external capabilities